

 [Bookmark this page](#)

 Calculator

General solution in terms of fundamental matrix



▶ 9:40 / 9:40

▶ 2.0x

🔊

🔍

CC

🗣️

Video
📄 [Download video file](#)

Transcripts
📄 [Download SubRip \(.srt\) file](#)
📄 [Download Text \(.txt\) file](#)

by an arbitrary square matrix of constants,
except you want to be sure that the determinant is not 0.
Well the determinant of this guy won't be 0,
so all you got to do is make sure that the determinant of C
isn't 0 either.
So in other words, the fundamental matrix
isn't unique.
But once you've found one, all the other ones
are found by multiplying it on the right
by an arbitrary square matrix of constants,
which is non singular, has determinants
not 0 in other words.

Suppose $\mathbf{x}_1, \dots, \mathbf{x}_n$ are n linearly independent solutions to the $n \times n$ system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, then the general solution is

$$\mathbf{x}(t) = c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n = \begin{pmatrix} | & & | \\ \mathbf{x}_1 & \cdots & \mathbf{x}_n \\ | & & | \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}. \tag{4.43}$$

Conclusion: If $\mathbf{X}(t)$ is a fundamental matrix, then the general solution is the product

$$\mathbf{x}(t) = \mathbf{X}(t)\mathbf{c}, \tag{4.44}$$

where $\mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$ ranges over all constant vectors.

Question 8.1 What do all fundamental matrices look like?

Recall that

1. each column of a fundamental matrix is a solution,
2. given a fundamental matrix $\mathbf{X}$, any solution is of the form $\mathbf{X}\mathbf{c}$ for some constant vector $\mathbf{c}$.

This means that all other fundamental matrices must be of the following form:

$$\begin{pmatrix} | & | & & | \\ \mathbf{X}\mathbf{c}_1 & \mathbf{X}\mathbf{c}_2 & \cdots & \mathbf{X}\mathbf{c}_n \\ | & | & & | \end{pmatrix} = \mathbf{X}\mathbf{C} \tag{4.45}$$

where $\mathbf{C}$ is the $n \times n$ matrix whose columns are the vectors $\mathbf{c}_i$:

$$\mathbf{C} = \begin{pmatrix} | & | & & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \\ | & | & & | \end{pmatrix} \tag{4.46}$$

and to ensure that the columns $\mathbf{X}\mathbf{c}_i$ are linearly independent, we need $\mathbf{C}$ to be invertible, i.e. $|\mathbf{C}| \neq 0$.

Conclusion: If $\mathbf{X}(t)$ is a fundamental matrix, then all other fundamental matrices are of the form

$$\mathbf{X}\mathbf{C} \quad \text{where } \mathbf{C} \text{ is } n \times n, \text{ and } |\mathbf{C}| \neq 0. \tag{4.47}$$

Remark 8.2 For the system on the previous page, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$, the general solution is

$$\mathbf{x}(t) = c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2 \quad \text{where } \mathbf{x}_1 = e^{2t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \tag{4.48}$$

$$\mathbf{x}_2 = e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ are the normal modes.} \tag{4.49}$$

This can be written in terms of the fundamental matrix $\mathbf{X} = (\mathbf{x}_1 \quad \mathbf{x}_2)$, as

$$\mathbf{x}(t) = \mathbf{X}\mathbf{c} = \begin{pmatrix} 2e^{2t} & e^{3t} \\ e^{2t} & e^{3t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}. \tag{4.50}$$

Most other fundamental matrices are “ugly.” For example, here is another one: $(\mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_2)$. We will use the simplest one, the one built from the normal modes, as much as we can.

Initial conditions

Once we have a fundamental matrix $\mathbf{X}(t)$, finding the solution $\mathbf{x}(t)$ with initial value $\mathbf{x}(a)$ at $t = a$ means finding the vector $\mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$ satisfying

$$\mathbf{X}(a) \mathbf{c} = \mathbf{x}(a). \tag{4.51}$$

Since $\mathbf{X}(a)$ is invertible for any a , we have

$$\mathbf{c} = \mathbf{X}(a)^{-1} \mathbf{x}(a). \tag{4.52}$$

Example 8.3 For the same system as above, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$, use the simplest fundamental matrix $\mathbf{X}$

$$\mathbf{X}(t) = \begin{pmatrix} 2e^{2t} & e^{3t} \\ e^{2t} & e^{3t} \end{pmatrix}.$$

to find the solution to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ satisfying the initial condition $\mathbf{x}(0) = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$.

Solution:

The solution to the initial value problem is $\mathbf{X}(t) \mathbf{c}$ for some constant vector $\mathbf{c}$. Thus

$$\mathbf{x} = \begin{pmatrix} 2e^{2t} & e^{3t} \\ e^{2t} & e^{3t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

for some c_1, c_2 to be determined. Set $t = 0$ and use the initial condition to get

$$\begin{pmatrix} 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

Solving leads to $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 6 \end{pmatrix}$. Therefore,

$$\mathbf{x} = \begin{pmatrix} 2e^{2t} & e^{3t} \\ e^{2t} & e^{3t} \end{pmatrix} \begin{pmatrix} -1 \\ 6 \end{pmatrix} = (-1) \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{2t} + 6 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t}.$$

Initial condition practice

2/2 points (graded)
In the example above, we found the general solution in terms of the fundamental matrix

$$\mathbf{X}(t) = \begin{pmatrix} 2e^{2t} & e^{3t} \\ e^{2t} & e^{3t} \end{pmatrix}$$

given by

$$\mathbf{x}(t) = \begin{pmatrix} 2e^{2t} & e^{3t} \\ e^{2t} & e^{3t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

Find coefficients c_1 and c_2 for the initial condition $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

-1

The initial condition gives the equation

© MITx Online. All rights reserved except where noted.

Accessibility.

You have used 1 of 3 attempts

Hide Discussion

Topic: Unit 3: Solving systems of first order ODEs using matrix methods / 8. General solution in terms of fundamental matrix

Show all posts ▼

by recent activity ▼

x